Operations	Fill in the blanks in the table given below of properties of Integers. (If a, b, c are integers, then $-a, -b, -c$ are also integers)					
	Closure	**Commutative**	**Associative**	**Identity**	**Inverse**	**Distributive**
Addition	$a+b$ is in $\mathbb{Z}$ E.g. $5+(-3)=2$ $\Rightarrow 2$ is in $\mathbb{Z}$	$a+b=b+a$ E.g. $5+(-3)=(-3)+5$ $\Rightarrow 2=2$	$(a+b)+c$ $=a+(b+c)$ E.g. $(2+3)+(-4)=1$ $2+[3+(-4)]=1$	$a+0$ $=0+a=a$ E.g. $(-4)+0$ $=0+(-4)=-4$	$a+(-a)$ $=(-a)+a=0$ E.g. $5+(-5)$ $=(-5)+5=0$	$a\times(b+c)$ $=(a\times b)+(a\times c)$ E.g. $2\times[3+(-5)]=-4$ $(2\times 3)+[2\times(-5)]$ $=-4$
Multiplication	ab is in $\mathbb{Z}$ E.g._____	$a\times b=b\times a$ E.g._____	$(a\times b)\times c$ $=a\times(b\times c)$ E.g. $(2\times 3)\times(-6)=-36$ $2\times[3\times(-6)]=-36$	$a\times 1$ $=1\times a=a$ E.g._____	Does not exist	Not Applicable
Subtraction	$a-b$ is in $\mathbb{Z}$ E.g._____	Fails $a-b\neq b-a$ E.g._____	Fails $(a-b)-c$ $\neq a-(b-a)$ E.g._____	Fails $a-0\neq 0-a$ E.g. $5-0=5$ $0-5=-5$ $5\neq -5$	Fails $a-(-a)$ $\neq(-a)-a$ E.g. $2-(-2)=4$ $(-2)-2=-4$ $4\neq -4$	$a\times(b-c)$ $=(a\times b)-(a\times c)$ E.g._____
Division	Fails $a\div b$ is not in $\mathbb{Z}$ E.g. $3\div 5=\frac{3}{5}$ does not belong to $\mathbb{Z}$	Fails	Fails	Fails	Fails	Not applicable

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